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Experiment 2

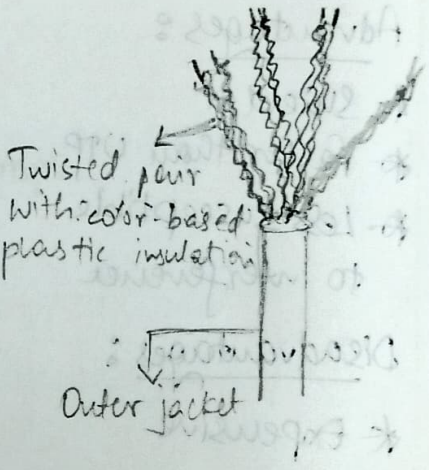
AIM: Study of different types of Network cable.

Different types of network cable:

1. Unshielded Twisted Pair (UTP) Cable
2. Shielded Twisted Pair (STP) Cable
3. Coaxial cable
4. Fibre Optic Cable

1) **UTP:**

Category 3 – 10 bps
Category 5 – 100 Mbps
Category 5e – 1 Gbps.



2) **Characteristics:**

Fast Ethernet, Gigabit Ethernet

Advantages:

→ cost effective
→ Easy to install.

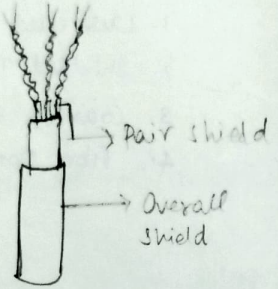
Disadvantages:

→ More prone to Electromagnetic Interference.

2) STP

Category 6, 6a. – 10Gbps.

Category 7 (SSTP) – 10Gbps.



Characteristics:
* Gigabit Ethernet, 10G Ethernet
* Widely used in data centres.

Advantages:
* Shielded
* Faster than UTP
* Less susceptible to interference

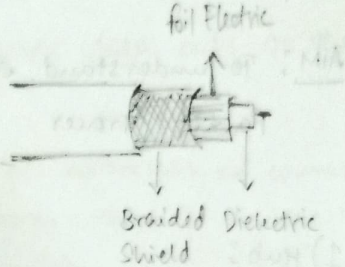
Disadvantages:
* Expensive
* Greater install effort

3) Coaxial Cable

RG-6, RG-59, – 10-100 Mbps.
RG-11

Characteristics
* Speed of signal is 500m.
* Television networks.
* High speed internet connections.

Advantages:
* High bandwidth
* Immune to interference
* Low loss of bandwidth.



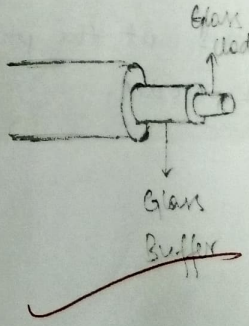
Disadvantages:
* Limited distance
* Cost

4) fibre-optic cable

Single & multi mode – 100 Gbps.

Characteristics:
* Max distance of fibre optic cable is 100 m.

Advantages:
* High speed.
* High bandwidth
* Secure



Disadvantages:
* Expensive
* Requires skilled installation.